AJKD Atlas of Renal Pathology: C1q Nephropathy

Agnes B. Fogo, MD,[1] Mark A. Lusco, MD,[1] Behzad Najafian, MD,[2] and Charles E. Alpers, MD[2]

Clinical and Pathologic Features

C1q nephropathy encompasses a range of biopsy findings that share deposition of dominant C1q, with varied clinical presentation. Patients are usually children or young adults who present with marked proteinuria, but may also have active urine sediment with red blood cells. Those with nephrotic syndrome typically have nonspecific light microscopy findings, with extensive foot process effacement and mesangial deposit shown by electron microscopy, identified as C1q by immunofluorescence. About 77% of these respond to immunosuppression. In contrast, patients with sclerosing or proliferative lesions revealed by light microscopy, the latter associated with subendothelial deposits, have worse prognosis, with stable kidney disease in about half of those with proliferative lesions.

Light microscopy: No specific lesions, or segmental sclerosis or endocapillary proliferation.

Immunofluorescence microscopy: C1q dominant mesangial deposits, which may extend to subendothelial areas.

Electron microscopy: Extensive foot process effacement and mesangial deposits in those with unremarkable light microscopic findings, or lesser foot process effacement and subendothelial extension of deposits in those with proliferative lesions. There are no reticular aggregates.

Etiology/Pathogenesis

The etiology of C1q nephropathy has not been defined. This lesion likely, in our opinion, may reflect a spectrum of disorders, with those without proliferation and with extensive foot process effacement more related to the spectrum of minimal change disease and focal segmental glomerulosclerosis, whereas those with proliferation are more akin to immune complex processes. The C1q deposits suggest an abnormality, not yet defined, of complement regulation.

Differential Diagnosis

The presence of deposits shown by electron microscopy, with dominant C1q staining, rule out minimal change disease and primary focal segmental glomerulosclerosis. Deposits with C1q could suggest lupus nephritis. However, the absence of immunoglobulin G dominant/codominant deposits or reticular aggregates, and lack of clinical history of systemic lupus erythematosus, are against this diagnosis.

Key Diagnostic Features

- C1q staining and mesangial deposits shown by electron microscopy
- Extensive foot process effacement in cases without proliferative lesions
- Lesser foot process effacement in cases with proliferative lesions
- No reticular aggregates, and lack of immunoglobulin G dominant deposits

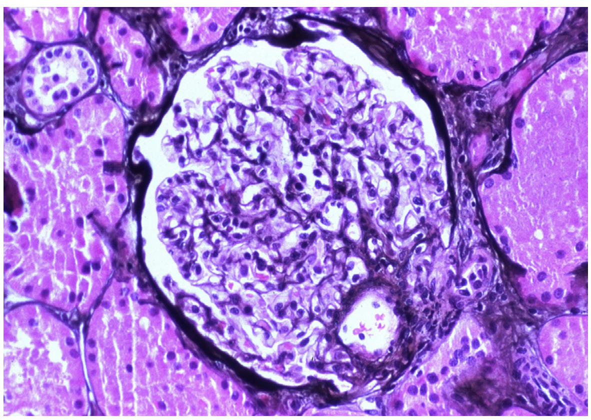

Figure 1. C1q nephropathy with mild mesangial hypercellularity with minimal increase in mesangial matrix (Jones silver stain). Reproduced with permission from *AJKD* 33(5):e1.

From the [1]Department of Pathology, Microbiology and Immunology, Vanderbilt University, Nashville, TN; and [2]Department of Pathology, University of Washington, Seattle, WA.

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Address correspondence to agnes.fogo@vanderbilt.edu

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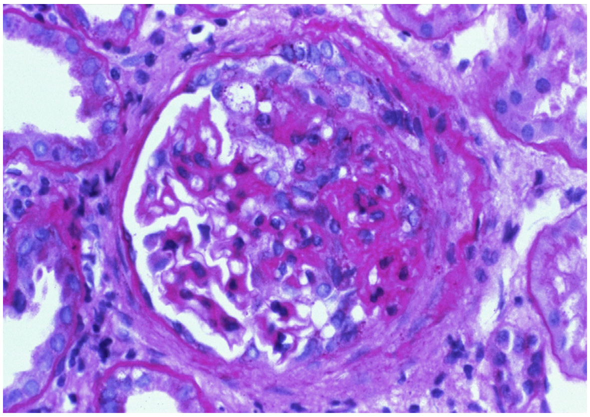

Figure 2. C1q nephropathy with segmental sclerosis with increased mesangial matrix and obliteration of capillary lumens and an adhesion to Bowman capsule (right half of tuft). The open left portion of the glomerulus shows mild to moderate increase in mesangial cellularity (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 33(5):e1.

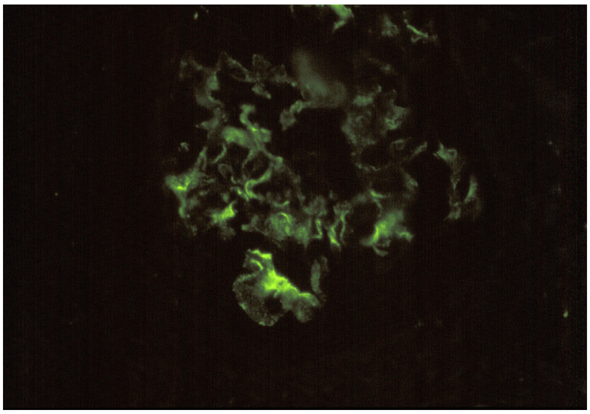

Figure 3. C1q nephropathy with mesangial staining for C1q, with segmental capillary wall extension. In C1q nephropathy there is lesser intensity staining for immunoglobulin and C3 than for C1q (immunofluorescence microscopy, C1q). Reproduced with permission from *AJKD* 33(5):e1.

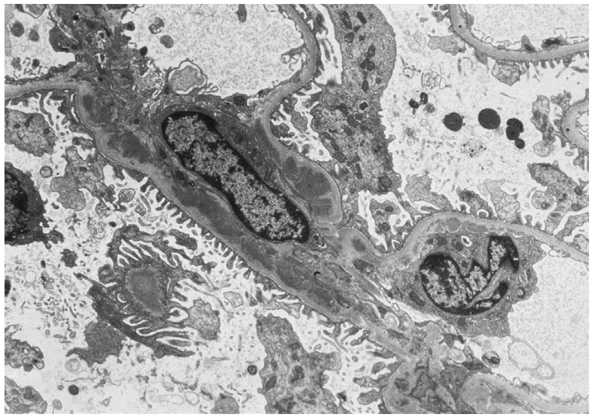

Figure 4. C1q nephropathy with mesangial deposits underlying the basement membrane as it traverses over the mesangial area, with subtotal overlying foot process effacement (electron microscopy). Reproduced with permission from *AJKD* 33(5):e1.